Hotel Booking Cancellation Prediction

A machine learning approach to predicting whether a hotel reservation will be cancelled

Project Report

Data understanding, cleaning and exploratory analysis of the Hotel Booking Demand dataset, with the proposed prediction and risk-scoring system.

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Dataset	Hotel Booking Demand (hotel_bookings.csv), 119,390 bookings
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Abstract

Hotel cancellations are costly. A room that was expected to be occupied can end up empty at short notice, which means lost revenue, wasted staffing and inventory, and rushed attempts to resell the room. This project builds the foundation of a machine learning system that analyses customer and booking details, such as lead time, deposit type, market segment, length of stay and special requests, to predict whether a reservation is likely to be cancelled.

The work uses the public Hotel Booking Demand dataset, which contains 119,390 bookings from a city hotel and a resort hotel, described by 32 attributes. The data is cleaned (31,994 exact duplicate rows removed, sparse columns dropped, remaining gaps filled), two features are engineered, and an exploratory analysis is carried out to find the booking patterns that separate cancelled from non-cancelled reservations. The analysis shows clear signals: cancellations are more common at the city hotel, peak in the summer arrival months, rise with longer lead times, and are far lower for guests who make special requests or need parking.

Project status. The Colab notebook covers data loading, cleaning, feature creation and exploratory analysis, and saves a cleaned dataset ready for modelling. Model training and evaluation are the next stage. Section 8 therefore describes the planned modelling approach, and this report does not quote accuracy or other model scores because none have been produced yet.

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1. Introduction and problem statement

Booking cancellations are one of the biggest sources of uncertainty in the hotel industry. Online travel agencies and flexible booking policies make it easy for guests to reserve a room and just as easy to cancel it. When a cancellation happens close to the arrival date, the hotel may be unable to resell the room, so the revenue is lost. To protect themselves, many hotels overbook, which can then backfire when too many guests turn up.

If a hotel could estimate in advance which reservations are likely to be cancelled, it could plan staffing and inventory more accurately, apply targeted policies such as deposits or reconfirmation calls, and manage overbooking with more confidence. This is a **binary classification** problem: given the details of a booking, predict whether it will end in cancellation (1) or not (0).

2. Objectives

- Understand the structure, quality and limitations of the hotel booking data.
- Clean the data so that it is reliable enough for machine learning.
- Explore which customer and booking characteristics are associated with cancellation.
- Prepare a cleaned dataset for training a cancellation prediction model.
- Define a prediction system that can convert model output into practical Low, Medium and High cancellation risk levels for hotel staff.

3. Dataset description

The project uses the Hotel Booking Demand dataset (*hotel_bookings.csv*). Each row is one booking made at either a City Hotel or a Resort Hotel. Arrival dates in the data span 2015 to 2017. The target column is **is_canceled**, where 0 means the booking was kept and 1 means it was cancelled.

Property	Value
Rows (original)	119,390 bookings
Columns (original)	32 (16 integer, 4 decimal, 12 text)
Hotel types	2 (City Hotel, Resort Hotel)
Target variable	is_canceled (0 = not cancelled, 1 = cancelled)
Not cancelled	75,166 bookings (62.96%)
Cancelled	44,224 bookings (37.04%)

The attributes fall into a few natural groups:

Group	Columns
Booking details	hotel, lead_time, deposit_type, booking_changes, days_in_waiting_list, distribution_channel, market_segment, reserved_room_type, assigned_room_type
Arrival date	arrival_date_year, arrival_date_month, arrival_date_week_number, arrival_date_day_of_month

Group	Columns
Stay and guests	stays_in_weekend_nights, stays_in_week_nights, adults, children, babies, meal, country
Customer history	is_repeated_guest, previous_cancellations, previous_bookings_not_canceled, customer_type
Price and requests	adr (average daily rate), required_car_parking_spaces, total_of_special_requests
Agent / company	agent, company (ID codes of the booking agent and company)
Outcome-related	reservation_status, reservation_status_date, is_canceled

The summary statistics show a typical booking is made about 104 days ahead on average (median 69 days, maximum 737), lasts roughly three and a half nights, and involves about two adults. The average daily rate has a mean of about 102 but a maximum of 5,400, which indicates extreme outliers.

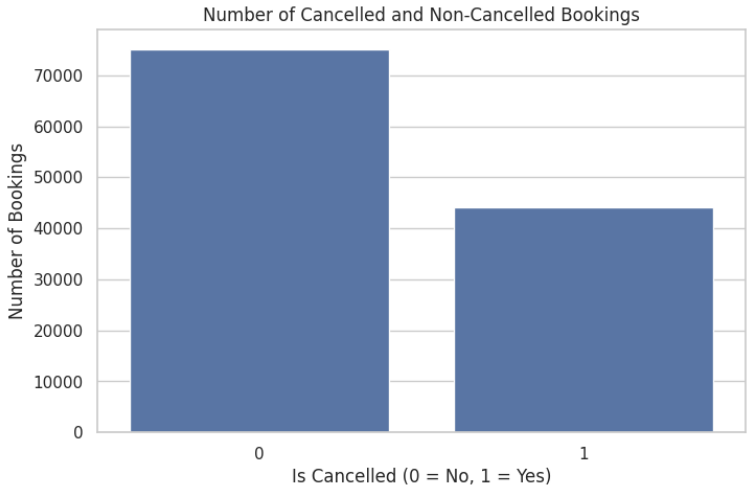


Figure 1. Number of cancelled and non-cancelled bookings in the original data. Roughly 37% of bookings are cancelled, so the classes are moderately imbalanced.

4. Data quality checks and cleaning

Before any analysis, the notebook checks missing values, duplicates and unusual records. The findings and the action taken for each are summarised below.

Issue found	Detail	Action taken
Missing: company	112,593 missing (94.31%)	Column dropped
Missing: agent	16,340 missing (13.69%)	Column dropped
Missing: country	488 missing (0.41%)	Filled with the label 'Unknown'
Missing: children	4 missing	Filled with the median (0)
Exact duplicate rows	31,994 duplicates	Removed, leaving 87,396 rows
Zero-guest bookings	166 records	Flagged, kept for now
Zero-night stays	651 records	Flagged, kept for now
Negative ADR	1 record	Flagged, kept for now

Why drop company and agent? The company column is missing for almost 94% of bookings, so it carries too little usable information. Agent is missing for about 14% of rows and has 333 distinct IDs, which would add a large number of sparse categories. Both were dropped to keep the first version of the dataset simple.

Why label missing countries as Unknown? Filling them with the most common country would invent information. A separate 'Unknown' category keeps the fact that the value was missing.

Unusual records. The zero-guest, zero-night and negative-rate records were checked but not deleted automatically. Unusual values are not always errors, so the notebook leaves them for a deliberate decision at the modelling stage.

After cleaning, the dataset has 87,396 rows and no missing values. The notebook reports that 26 duplicate rows appear again once the agent and company columns are removed, because rows that differed only in those columns now look identical. This is a small number, and removing them is a simple improvement for a later iteration.

5. Feature engineering

Two simple features were added to make booking patterns easier to study:

- **total_nights** = stays_in_weekend_nights + stays_in_week_nights, the full length of the stay.
- **total_guests** = adults + children + babies, the party size.

The arrival month was also converted into an ordered category (January to December) so that month-based charts appear in calendar order. The final cleaned dataset has 87,396 rows and 32 columns and was saved as *hotel_bookings_cleaned.csv* for modelling.

6. Exploratory data analysis

All charts below come from the cleaned data. Because duplicates were removed, the overall cancellation rate in the cleaned data is lower than the 37% seen in the original file; the charts should be read as relative comparisons between groups.

6.1 Hotel type and season

Cancellation is clearly higher at the City Hotel (about 30%) than at the Resort Hotel (about 23%). Seasonality is also visible: the cancellation rate is highest for arrivals in July and August (above 31%), is also elevated in April and June, and is lowest for November arrivals (about 21%).



Figure 2. Cancellation rate by hotel type.

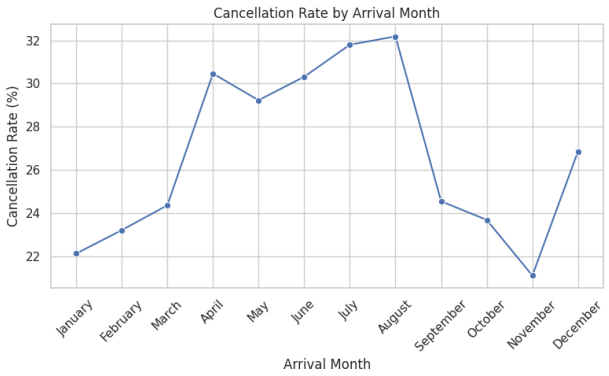


Figure 3. Cancellation rate by arrival month.

6.2 Deposit type and market segment

The deposit chart shows a surprising result: bookings with a **Non Refund** deposit have a cancellation rate of about 95%, far above No Deposit (about 27%) and Refundable (about 24%). This is the opposite of what one would expect, since guests who pay a non-refundable deposit have the most to lose. A likely explanation is that these bookings came mainly from certain channels and were cancelled in bulk, and it should be investigated rather than assumed. It also makes deposit_type a very strong feature for any model.

By market segment, Online Travel Agent (TA) bookings have the highest cancellation rate among the large segments (about 35%), followed by Groups (about 27%). Corporate, Complementary and Direct bookings are the most reliable, at roughly 12% to 15%. The 'Undefined' segment shows a rate near 100%, but it contains very few bookings, so this value should be treated with caution.

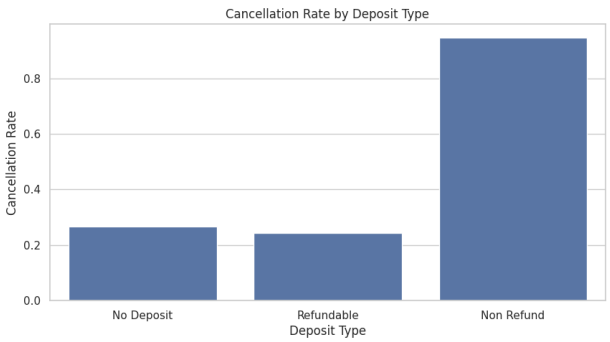


Figure 4. Cancellation rate by deposit type.

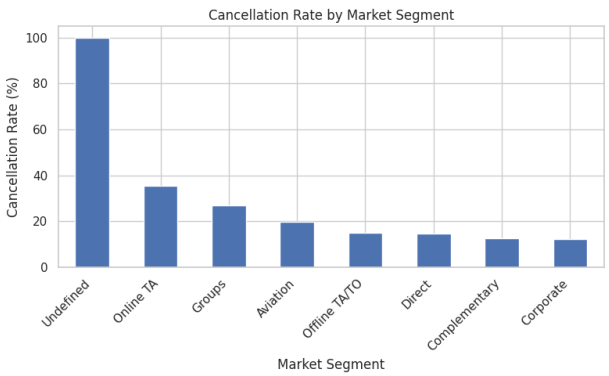


Figure 5. Cancellation rate by market segment.

6.3 Lead time and price

Lead time, the number of days between booking and arrival, is a key signal. Cancelled bookings have a higher median lead time (roughly 80 days) than kept bookings (roughly 35 days), and the longest lead times, above 300 days, are much more common among cancellations. The earlier a guest books, the more time there is for plans to change.

Average daily rate (ADR) is also higher for cancelled bookings in both hotel types, and the gap is wider at the Resort Hotel. ADR was plotted up to the 99th percentile so that extreme values such as the 5,400 maximum do not hide the main pattern.

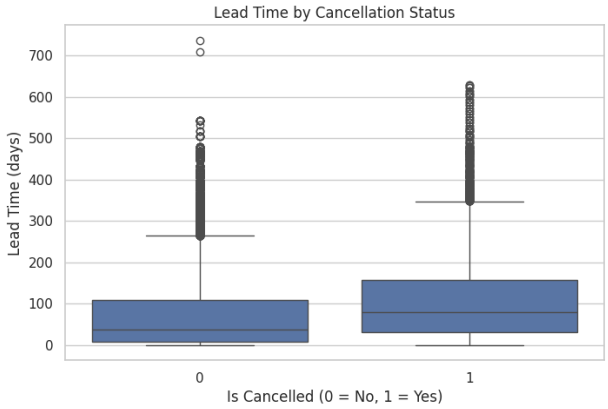


Figure 6. Lead time by cancellation status.

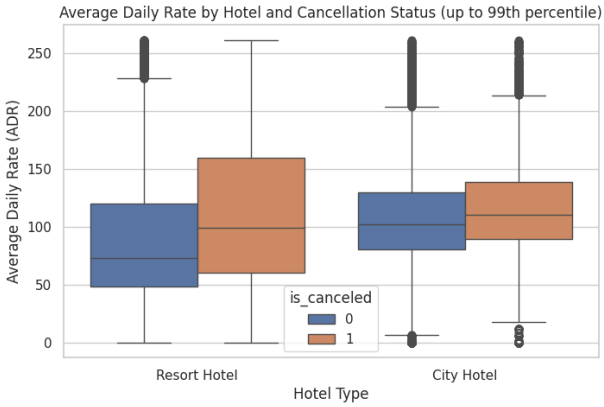


Figure 7. ADR by hotel type and cancellation status.

6.4 Guest behaviour

Guests who make special requests are less likely to cancel: kept bookings average about 0.76 special requests, against about 0.53 for cancelled ones. The pattern for parking is even starker, with kept bookings requiring about 0.12 parking spaces on average and cancelled bookings requiring almost none. Both behaviours suggest commitment: a guest who has arranged parking or made special requests has already invested effort in the trip. Cancelled bookings also have a slightly larger average party size (about 2.15 against 1.97 guests).

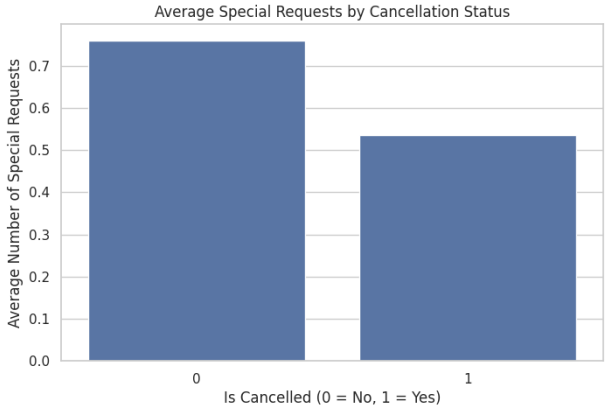


Figure 8. Average special requests by cancellation status.

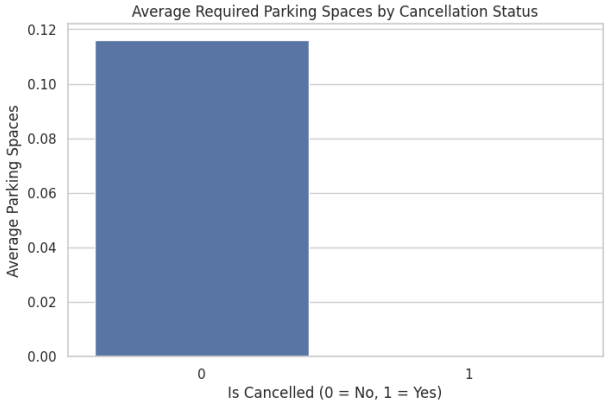


Figure 9. Average parking spaces required by cancellation status.

6.5 Correlation between numeric features

A correlation heatmap of the numeric features was produced to check how the variables relate to each other and to the target, and to spot features that carry overlapping information. It is a useful first

screen, though correlation only captures linear relationships and will miss patterns in categorical columns such as deposit type or market segment.

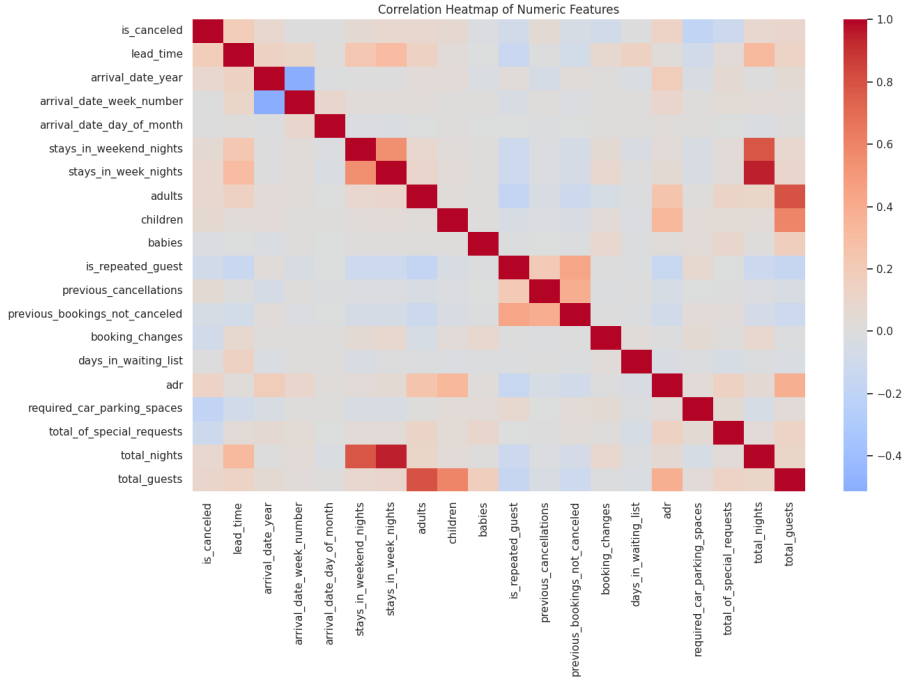


Figure 10. Correlation heatmap of numeric features.

Summary of key findings

Factor	Observation from the data
Hotel type	City Hotel cancels more (about 30%) than Resort Hotel (about 23%)
Arrival month	Highest in July and August, lowest in November
Deposit type	Non Refund bookings cancel at about 95%, which needs investigation
Market segment	Online TA and Groups cancel most; Corporate and Direct cancel least
Lead time	Cancelled bookings are made earlier (higher median lead time)
Special requests	More requests goes with fewer cancellations
Parking	Bookings requiring parking are rarely cancelled

7. Important considerations before modelling

Data leakage. The dataset includes *reservation_status* (Check-Out, Canceled or No-Show) and *reservation_status_date*. These record what happened to the booking *after* the outcome was known, so reservation_status effectively contains the answer. They must be excluded from the model's input features; otherwise the model would look almost perfect in testing and be useless in real use, where the outcome is not yet known. The same care applies to assigned_room_type, which may only be fixed close to arrival.

Class imbalance. Roughly one booking in three or four is cancelled. Accuracy alone can be misleading in this situation, so precision, recall and F1-score should be reported alongside it.

Unusual values and outliers. Zero-guest bookings, zero-night stays, a negative ADR and extreme ADR and lead-time values should be reviewed and handled deliberately, for example by removing clear errors or capping extremes.

Categorical features. Columns such as hotel, market_segment, deposit_type, customer_type, meal and country are text and need encoding (for example one-hot encoding) before most models can use them. Country has 177 distinct values and may need grouping.

8. Proposed modelling approach

This section describes the planned next stage. It has not been run in the current notebook, so no results are reported here.

Using the cleaned dataset, the plan is to:

- **Select and encode features.** Use the booking, customer and stay attributes, exclude the leakage columns, and encode categorical variables.
- **Split the data** into training and test sets using a stratified split so that both keep the same cancelled / not-cancelled proportions.
- **Train a baseline model.** Logistic Regression gives a simple, interpretable starting point and shows which features push the prediction towards cancellation.

- **Train a stronger model.** A Random Forest classifier can capture non-linear relationships and interactions between features, and also provides feature-importance scores.
- **Evaluate both models** on the unseen test set using accuracy, precision, recall, F1-score, ROC-AUC and the confusion matrix, and compare them.
- **Convert probabilities into risk levels.** The model outputs a probability of cancellation for each booking. Splitting this into Low, Medium and High risk bands, with thresholds chosen from the evaluation results, makes the output easy for hotel staff to act on.
- **Deliver through a simple interface.** A Streamlit dashboard where a user enters booking details to see the predicted risk, together with analytics views of the patterns found in this report.

Stage	Input	Output
Data preparation	Raw hotel_bookings.csv	Cleaned dataset (done)
Exploration	Cleaned dataset	Patterns linked to cancellation (done)
Model training	Encoded features, training split	Logistic Regression and Random Forest models (planned)
Evaluation	Test split	Metric comparison (planned)
Risk scoring and interface	Booking details	Cancellation probability and risk level (planned)

9. Business value

A working cancellation predictor could support several decisions at a hotel:

- **Revenue protection.** Flagging high-risk bookings early gives the hotel time to reconfirm with the guest or resell the room before it goes unused.
- **Smarter overbooking.** Instead of using a fixed overbooking percentage, a hotel can base it on the actual predicted cancellations for each date.
- **Resource planning.** Staffing, housekeeping and food and beverage purchasing can be matched to expected occupancy rather than booked occupancy.
- **Targeted policies.** High-risk segments or lead-time ranges can be offered deposits or stricter terms, while low-risk guests keep a flexible experience.
- **Channel decisions.** The segment-level differences help hotels judge how reliable each booking channel is.

10. Limitations, future work and conclusion

Limitations

- The model stage is not yet complete, so predictive performance is still unknown.
- The data covers only two hotels in 2015 to 2017. Patterns may not carry over to other hotels, regions or the present day.
- The EDA shows association, not cause. For instance, the high cancellation rate for Non Refund deposits needs further investigation before it is used in any policy.
- Dropping the agent and company columns removes some potentially informative detail.

Future work

- Train and tune the Logistic Regression and Random Forest models and compare them, then try gradient boosting methods.
- Handle the unusual records and the 26 reappearing duplicates, and test whether re-adding agent as a grouped feature helps.
- Address class imbalance (class weights or resampling) and choose thresholds that reflect the cost of a missed cancellation.
- Build and deploy the Streamlit dashboard for prediction and analytics.

Conclusion

This project has laid the groundwork for a hotel booking cancellation prediction system. The raw data of 119,390 bookings was cleaned to 87,396 reliable records, enriched with two engineered features, and explored in detail. The analysis shows that cancellation is strongly tied to hotel type, season, deposit type, booking channel, lead time and guest behaviour, which gives good reason to expect that a well-built model can predict cancellations usefully. With the leakage columns excluded and the modelling steps in Section 8 completed, the system can turn these patterns into booking-level risk scores that help hotels reduce revenue loss and plan more effectively.